

12. (1.75 points)

a. (1 point)

When there are negative incremental values in loss development data, the log-link function used in the GLM framework will fail to yield values usable to parameterize the model. Describe two modifications to the log-link function that address this issue.

b. (0.75 point)

Given the following cumulative development triangle of reported losses to be used in a GLM bootstrapping model:

Accident	12	24	36	48
<u>Year</u>	<u>Months</u>	<u>Months</u>	<u>Months</u>	<u>Months</u>
2011	1,500	1,300	1,200	1,250
2012	2,000	1,500	1,750	
2013	1,750	2,000		
2014	2,200			

Explain which of the two methods identified in part a. above is the more appropriate method to use on this data set.

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QUESTION 12				
TOTAL POINT VALUE: 1.75	LEARNING OBJECTIVE: A8: Identify data issues and related model adjustments for reserving models.			
SAMPLE ANSWERS				
Part a: 1 point				
Sample Answer 1				
1. When column sum is positive:				
$\ln(q(w,d)) = \begin{matrix} \ln(q(w,d)) & \text{for } q(w,d) > 0 \\ 0 & \text{for } q(w,d) = 0 \\ -\ln(-q(w,d)) & \text{for } q(w,d) < 0 \end{matrix}$				
2. When column sum is negative, subtract the largest negative from each incremental loss, and fit GLM using the modified triangle. Adjust the fitted incremental loss back by adding the same (largest negative) to the fitted incremental loss.				
Sample Answer 2				
1. Use $-\ln(-\text{loss})$ if the incremental loss is negative.				
2. Add constant to each cell in the triangle so that each cell is positive. After GLM estimation, deduct the constant from each cell.				
Part b: 0.75 point				
Sample Answer 1				
Incremental Loss Triangle				
AY	12	24	36	48
2011	1500	-200	-100	50
2012	2000	-500	250	
2013	1750	250		
2014	2200			
Column sum of age 24 is negative. The second model is more appropriate as GLM will fail with negative column sum.				
Sample Answer 2				
More appropriate to use method 1(subtract the largest negative value to each cell in the triangle and solve for the parameters to get the fitted mean and then add the same value for each cell) since column sum at 24 months is negative. In this case, subtract -500 from each cell. This ensures all columns sums are positive so a solution can be found.				
EXAMINER'S REPORT				
In general, candidates performed fairly well on this question, though it was not common to earn full credit.				
Part a				
Candidates were expected to know the two adjustments and most of the time they were				

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successful at it. However, they at times had difficulty fully explaining all the steps of each adjustment.

Part b

Candidates were expected to identify which of the two adjustments from part a. would address the data issue in the data presented here, and then to explain why it was the more appropriate adjustment to use.

In most cases candidates identified the correct adjustment. However, the majority omitted an explanation for why this particular adjustment was necessary (GLM would not find a solution).